

Googling the Archive: Digital Tools and the Practice of History

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This article argues the digital tools and search environments that increasingly support historical scholarship in rhetoric and composition have material and epistemological implications for how we discover, access, and make sense of the past. In light of these changes, I suggest that more explicit reflection and discipline-specific conversation around the uses and shaping effects of these technologies is needed. Tracing my own digitally enabled search for information about an early-twentieth-century advice writer named Frances Maule, I describe how mass digitization has shifted conditions of findability. I conclude by outlining a heuristic for critical reflection—a “principle of proximity”—and urging rhetoric and composition historians to take a more active role in shaping the emerging landscape of digital research.

[W]e must develop new tools, because our old tools were not designed for the challenges we now face.

—Richard Enos

The history of knowledge is the history of forms of expression of knowledge, and those forms change.

—Johanna Drucker

Writing in 1997, Linda Ferreira-Buckley described a scholarly meeting she had attended at Cambridge University several years earlier as “exhilarating

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and frightening.” It was exhilarating, she explained, because it was evident that “new research tools (such as databases from publishing records)” were enabling “historical recovery of a kind never before possible.” The frightening and humbling part was the sheer volume of information to be processed and made sense of—far beyond the capacity of an individual researcher (Ferreira-Buckley 1997, 27). Also writing in 1997—in this case to register a domain name—two Stanford graduate students named Larry Page and Sergey Brin chose the name “Google” for a search engine they had recently developed. As a reference to the mathematical term *googol* (the number 1 followed by 100 zeros), the name captured their intent to “organize a seemingly infinite amount of information on the web” (“Google History” 2011; see also Brin and Page 1998; Vise and Malseed 2005, 17, 39).

In the years since these writings, digital texts and technologies have become an ever more familiar feature of our methodological landscape and digitized information ever more abundant. The volume of digitized material available to historians continues to grow as entities like the HathiTrust (<http://www.hathitrust.org>) and Internet Archive (<http://www.archive.org>) pursue large-scale digitization as a means of preserving cultural memory and as Google aims to make history by scanning all of the world’s books—an estimated 129 million¹—within the next decade. By 2010, the Google Books project had scanned “more than 15 million books from more than 100 countries in over 400 languages” (Crawford 2010); in 2008, Microsoft reported that it had digitized 750,000 books and indexed 80 million journal articles through its (now retired) Live Search Books and Live Search Academic projects (Nadella 2008); and, in 2009, the Library of Congress’s Chronicling America newspaper database posted its one millionth page. And these are just a few of the more prominent digitization projects.

Given this growing abundance of digital information, we would do well to ask how digital environments promise to inform or transform our work as historians and what we can do to foster more explicit, discipline-specific conversations that consider the role and influence of digital technologies in our research. The digital tools and structures that increasingly support our research efforts have material and epistemological implications for how we discover, access, and make sense of the past—for the “methodological motion” (Murphy et al. 1988, 17) that propels a researcher’s inquiries down one path rather than another. In much the same way that the built environment of a building or city encourages or discourages certain kinds of physical movement, so too do built environments for historical research encourage or deter us from moving along certain paths of inquiry: unprocessed archives are harder to find; search engine results direct our gaze one way rather than another; library catalog entries make it easy to locate books by author name but less so to track historical documents whose author and title we do not yet know. Digital tools and digitized sources extend, remediate, and transform our research environments; these transformations, in turn,

open up possibilities for new kinds of interaction with, and new movements through, the archive. Digital environments thus have the potential to reorient us—both physically² and conceptually—to our research subjects and the artifacts through which we come to know them. The nature of that reorientation depends partly on us: whether we choose to be alert and active users and shapers of technology or are content to use these tools more or less as they come to us. My interest, then, is in the digital structures of the archive, our relationship to those structures, and our capacity to develop critical conversations around them—whether we’re talking about the back-end database, the front-end interface, or the built and improvised paths we take as we move through these digital spaces.

In what follows I consider mass digitization and digital technologies as they relate to researcher positionality and historical research processes. Specifically, I argue that digital tools mediate our discovery and interpretation of historical research subjects and thereby shape the histories that researchers find themselves more or less inclined (or able) to write. This shaping should not be mistaken for hard technological determinism, but is instead more akin to the agency or shaping influence that some theorists grant to objects and technologies—the way that “the material world pushes back on people because of its physical structure and design” (Johnson and Wetmore 2009, 151).³ If we are willing to grant that technologies inform a researcher’s inquiry and, thus, her relationship to her subject (whether because they constrain or support types of research, or because technologies, as Langdon Winner [1989] suggests, “have politics”) then we might be persuaded that cultivating a critical awareness of technology is important for theorizing researcher positionality more generally. Certainly, the move to acknowledge one’s position as a researcher—to describe one’s situatedness—has become, as Cheryl Glenn and Jessica Enoch note, “almost commonplace” (2010, 21). Yet as familiar as these positioning moves have come to be, they have not typically been concerned with explicating the role that digital technologies might play in positioning the historical researcher or mediating that researcher’s relationship to her subject.

The idea that rhetoric and composition scholars ought to give more attention to historical research methods is not a new one, but the literature in this area has remained thin until quite recently. Barbara L’Eplattenier observes as much in her 2009 opinion piece for *College English*, where she argues that we need to develop a more explicit literature of historical method in rhetoric and composition studies (68–69). That such a dearth exists is doubly true for discussions of method that take up emerging digital tools and environments. While a good deal has been written on digital technologies by scholars working in other specializations (digital humanities and computers and writing, for example),⁴ there has been little focused commentary in this vein from rhetoric and composition historians. There are signs that this may be changing: several essays dealing specifically with digital technology have

appeared in recent collections emphasizing archival research methods,⁵ as well as more that contain passing references to computer technologies of various kinds (for example, see essays by Warnick [2010] and Yakel [2010] in Ramsey et al. [2010]). Yet while such information is helpful (particularly for those new to historical research), more critically inflected work is needed as well: work that not only points us to relevant databases and explains how to conduct searches within them or navigate the deep Web, but also considers the structures of the digital tools themselves, and whose practices, values, and investments they represent. In much the same way that scholars have come to recognize the politics of “conventional” archives and begun to historicize and excavate their creation (Finnegan 2006, 118; Stoler 2002), we should be attuned to, and ready to critically engage, the production of “digital archives” in our own time.

I offer this article as a step toward such critical reflection; in it, I describe the shaping influence of mass digitization on my own research and raise questions about what these shifts in method might mean for research in the history of rhetoric and composition more generally. How might we respond as a field and as individual researchers? How do we make these digital mediations a more visible part of our process and therefore more available for reflection and examination? How might we as historical researchers take a more active and ambitious role in shaping these emerging technologies?

These are bigger questions than I can hope to answer here—and they are in any case questions that neither can nor should be answered by any one scholar. To begin to answer them requires that we bring to historiography’s digital environment the same values of reflection, attention to material conditions, and commitment to tracing sites and flows of power that already inform much existing rhetorical scholarship. One way to bring this critical scrutiny to bear is by sharing and circulating research narratives that make the processes of historical research visible to a wider audience—similar to what Gesa Kirsch and Liz Rohan have called “giving one’s game away” (2008, 1)⁶ but with particular attention to the mediating role of technologies. In what follows, I give away some of my own game by sharing a few technology reflections grounded in my current research project, which historicizes a body of career advice literature for the “business girl” that flourished in the United States in the early twentieth century, particularly in the years between World War I and World War II. I focus this discussion around my digitally enabled search for information about an advice writer named Frances Maule and describe how mass digitization shifted conditions of findability for this author; this shift affected the texture and subsequent direction of the search and my interpretation of her work. Following this discussion, I examine digital keyword searching and introduce “proximity” as a concept that articulates technological mediation as an element of researcher positionality. Finally, I conclude by raising questions about the roles we envision for technology as we look toward the future of historical research in rhetoric and composition in a culture of mass digitization.

FINDING FRANCES MAULE

My first encounter with Frances Maule did not take place in a digital environment but instead in the main bookstacks of the library at the University of Illinois at Urbana-Champaign. I was sitting on the cold, hard floor, paging through early issues of the American Business Writing Association's *Bulletin* (now *Business Communication Quarterly*) in hopes of finding information about the role of women in the development of business writing. I had been flipping through the newsletter's yellowed, typewritten pages for some time when I came across a bibliography of readings that an instructor named Mamie Meredith had used in her business writing courses at the University of Nebraska and had submitted to the *Bulletin* to be shared with other instructors (10). Published in 1944, the bibliography was presented as a helpful resource for business writing instructors, given "the preponderance of girls" enrolled in business writing classes at that time (likely due to the popularity of clerical work in general as well as the demand for trained office workers during World War II). Meredith's bibliography listed around thirty items, including four books by a writer named Frances Maule.⁶ Curious to know more about what sorts of books instructors were recommending to the women in their business writing courses, I examined a few of the most interesting-sounding titles, including Maule's. The texts on the list were a mix of popular career advice, self-improvement, and vocational guidance, and were geared toward helping women to choose a career and "fit" themselves for business.

This career advice literature eventually became the subject of my dissertation and then a book project, and I would end up reading advice texts that spanned a period of just over fifty years. Yet even as I engaged with advice from many more writers across a broader span of time, I continued to be drawn to Maule, and my curiosity translated into repeated online searches. Part of the attraction was her prose, which was lively and engaging: she gave good quote. Her vivid commentary and its emphasis on women's bodies—on "b.o." and halitosis, on frequent hair washings and what not to wear—seemed to confirm stereotypes of clerical work and gender roles during this period as well as the shallow prescriptiveness of etiquette and conduct guides more generally. Reminding readers that the bodily rhetorics of the businesswoman demanded emotional restraint, Maule instructed her readers: "Glad, sad or mad, office etiquette requires that you should always wear in the office a look of cheerful, good-humored impersonal composure and reserve. Emotion—any kind of emotion—is altogether out of place" (1934, 45). It was good stuff, in the way that cringeworthy advice from an earlier era can be considered "good" (good for showing how consumer products shaped discourses around women's bodies, for example), and my initial (cringing) reaction to such passages seemed consistent with that of other historians who quote from texts like Maule's

and who read such passages as evidence of an environment that imposed narrow gender roles. Few of the historians who cite Maule's advice texts have had much to say about Maule herself, however—in their interpretations she (as the author) is largely subordinated to her texts, which serve instead as “facts”: evidence of attitudes toward women and beliefs about their constrained roles in office culture (for an example of this, see Todd [1993, 305]). I nevertheless tried to read “resistantly” and to press past the literal meaning of the text on the page by learning as much as I could about Maule's life and the experiences she brought to her advice writing.

As I have researched Maule, I have been struck by the extent to which Web-based searches across newly available volumes of digitized content have made it possible to assemble a more detailed, textured account of her life. She was, at the time that I began my research, a “hidden in plain sight” subject: one who is present in an array of published materials, both primary and secondary, and yet whose presence in many of those materials is not visible through traditional library finding aids. Searching Amazon's full-text database and using Google Books proved to be an effective way to locate and cull secondary sources that discussed Maule or her work—in some cases only a footnote, but a footnote that pointed toward new information or a new primary source to track down. Similar types of searching turned up articles she had published as a freelance magazine writer and, as more finding aids came online, I identified more archives that hold personal documents and records that mention her. As this research unfolded, mass digitization was taking off (2004–2010), and new sources were steadily becoming available; the differences I saw in search results across time made me more aware of what digitization was making possible for my own research and how it was sustaining this particular line of inquiry.

At first, I knew little about Maule beyond what the library catalog and WorldCat told me: that she had authored eight career advice books between 1934 and 1961 and, two decades before, had written and edited materials for the National American Woman Suffrage Association (NAWSA) in the final push for suffrage. I kept searching, motivated in part by a desire to understand the “mentalities” and culture (Enos 2009, 40, 44) that informed Maule's writing of career advice for women. What worldview or ideological commitments might have accounted for both her suffrage work and her advice writing? Had she been among the more conservative members of the suffrage movement (those who saw suffrage as a way for women to more effectively exert their innately feminine qualities)? Had she simply grown more conservative with age? Or did she see career advice as an extension of that impulse (evident in her suffrage work) for enlarging women's opportunities? Though Maule was not the main focus of my project, these questions nagged. In the years that followed my first encounter with her in the library stacks, I continued to look for answers, gathering information bit by bit, mainly through

persistent online searching, which in turn led to more sources (online and in the physical library stacks). The results of these searches helped fuel my curiosity and keep this line of inquiry alive.

I looked for leads wherever I could think to do so, including searching commercial sites like Amazon and Flickr, and using Google's various tools (Book Search, Scholar, and the advanced search tool) as well as databases behind the library's digital walls, like ProQuest Historical Newspapers and the Gerritsen Collection (a database of women's history documents). I searched Internet Archive and, when HathiTrust came on the scene, I searched that too. Over time, Maule's name proved to be an effective search term; I repeated the same searches across a range of databases and, as more digitized content became available, gathered more new information. For example, by repeatedly using "frances maule" as a search term within Amazon.com and Google Books, I was able to quickly identify new scholarship soon after it was published that included passing mentions of Maule.

Thus, my forays into the digital archive were not in vain and, as the bits of information from primary and secondary sources accumulated, a more interesting and complicated picture emerged. I learned that in addition to being an advice writer and suffragist, Maule had also been a radical feminist, socialist, journalist and freelance writer, wife, publicist for the Henry Holt company (Scanlon 1995, 187), advertising copywriter, Heterodoxy club member,⁷ editor of *Independent Woman* magazine, and (according to one historian) a "probable lesbian" (Schwarz 1986, 36; Scanlon 1995, 189). This thickened description demanded a more nuanced reading of her texts, one that grappled with the contradictions and complexity of feminism in the early twentieth century and the challenges of selling new professional and workplace roles for American women. Maule, whom I had initially imagined to be a stereotypical reinforcer of conventionally gendered roles, traits, and behaviors, would not be so easily pigeonholed. "Selling" emerged as a theme that runs through her professional and political work, and I developed an appreciation for her understanding of bodily rhetorics and publicity tactics.

None of this necessarily recasts Maule as a heroine or rhetorical role model, but that is not really the point. The point, and the challenge, is to reconstitute as best we can the rhetorical situation at hand. The result of that effort, in this case, was to bring more interpretive dimension to what had otherwise been rather flat and generally arhetorical treatments of Maule's advice writing. Although it is not entirely inaccurate to read this research account as a kind of technological success story, I would be remiss if I didn't acknowledge that Maule's career as a published writer and her status as a white, well-educated woman with a social network of powerful women and men had something to do with why there were materials to be found and leads to be followed. Digital technology cannot rectify the

omissions or exclusions of the past, though it can, I think, help us make more of the traces that do remain and perhaps gives us means to knit those traces together more readily. One point to underscore here is the importance of historical work not simply for “recovering” women who have been neglected or shallowly represented by other historians but also for recognizing their rhetorical practices as such, placing those practices in context, and tracing them across the span of a life or career. With such tracings we create opportunities to further understand the transfer of rhetorical and literacy skills through time and across contexts: from one activist site to another, from school to work, from work to community and political life, and so on. The broadened digital archive and its constituent tools represent additional means by which to pursue these historiographical aims, augmenting more established archival methods.

Each historical project comes with its own twists and turns, which define and shape the final outcome: the history that gets written. When a historian arrives at a particular understanding, the resulting history, as Nan Johnson reminds us, “bears the unique mark of how the investigative method has proceeded and of the inevitable hierarchy of topics and conclusions that have emerged to guide conceptual development” (Murphy et al. 1988, 17). As we read research narratives from many scholars, however, there is also a sense of pattern and similarity that emerges in the form of repeated themes: serendipities, wrong turns or dead ends, *aha!* moments, or insights gained through a discovered document or visit to a physical site. Developing a sense of those patterns in digital environments—through discussions of methodology and sharing stories—is one way to make them more visible in our scholarly conversations. This means actively cultivating an awareness and dialogue around digital tools, and not treating digital search and discovery activities as transparent, neutral, or inconsequential to the acts of invention and interpretation that lie at the heart of rhetorical histories. As I argue in the conclusion, this type of critical dialogue is important for advancing our own disciplinary needs and interests at a moment when new search tools and environments are being developed. It means, as well, training graduate students (and ourselves) to be well-rounded historical researchers who make creative and thoughtful use of new tools, regarding them not as a one-for-one substitute for working with physical archives or documents printed on paper but as complementary resources. Indeed, one of the benefits of combining online and offline research may be (at least in the short term) a heightened awareness and deepened understanding of structures in *both* contexts. Whether the search takes place online or not, attention to mediation, materiality, and the position of the researcher are important; in the next two sections, I explore these elements and their relationship to research that uses digital tools.

STRUCTURES OF THE SEARCH

In the closing paragraphs of her instructive “Archival Survival” essay, Lynée Lewis Gaillet poses a series of questions, including the following: “How does the historian’s line of inquiry accommodate online searches?” and “[I]n what ways must we shift our method/methodological processes when researching historical documents/issues online” (2010, 38). These questions are well worth trying to answer; Gaillet’s first question, for example, serves to remind that a given line of inquiry might not be well suited to online searching. At the same time, it bears noting that in both questions (particularly the second) it is the researcher who is assumed to be accommodating herself to the technology. In fairness to Gaillet, I think many of us do, as a matter of practical necessity, make such accommodations (alas, the archive does not remake itself upon the historian’s command!) and sometimes those accommodations are generative. Still, there are at least two potential problems I see with the image of the researcher as technological accommodationist. The first is that it encourages a view of digital research tools and environments as something that scholars have little control over; the second is that by accepting this lack of control, we may fail to consider actions or practices that suggest otherwise. If we invert Gaillet’s questions, however, we might instead ask: How can *technologies be made to accommodate* the historian’s line of inquiry? In what ways might we imagine (and then advocate for) such technologies?

Understanding the degree to which particular technologies advance our historical inquiries (not to mention ethical and theoretical commitments) demands that we reflect on the technologies themselves and the ways their structures “push back” against our inquiries. In this section I consider the digital keyword search—the unit of action that has become so emblematic of research in an era of mass digitization and the means by which we discover, enter, and “invent” the archive. My interest in keyword searches is motivated partly by their importance for my own research but also by a concern that as these little white search boxes proliferate and mediate more and more of our research, they will effectively serve to “black-box,” or obscure, the underlying structures that enable their (and our) work.⁸

Let us return, then, to the two “writings” that open this essay (by Ferreira-Buckley and by Brin and Page, respectively), which represent two common and, I think, complementary ways that searches are experienced in digital environments. We can distinguish between these two types of experience in the following simplified way: on one hand, queries that search fixed *descriptions* of a document or artifact (by author, title, or subject heading, for example), and on the other, queries that directly search the *contents* of the document or artifact itself. The database of publishing records that prompted Ferreira-Buckley’s expression of awe was likely (and I am making

an educated guess here) an example of the former. Publishing records, like library catalog entries, are the sort of content that have typically been organized according to a fixed classification schemes developed and maintained by an expert. To search in this context, one must know or learn the expert's scheme (a list of established subject headings, for example). Brin and Page's Google search engine, by contrast, is designed to search the full content of the disorganized sprawl of the Web, and nevertheless returns results with a high degree of relevance. To search with Google, no special taxonomy is required: the system has been designed to learn from prior searches in order to handle new ones.

I think there is value in pausing to consider how these searches work, how they are structured. To think about how searches function (experientially as well as technically) is in some sense to work against the transparency and passive consumption that familiarity breeds. On a practical level, understanding something about how search technologies work is valuable for our research, for planning and building discipline- or project-specific research tools and platforms like the collaborative archival database that Glenn and Enoch have proposed (2010, 19–20) and for participating in broader discussions of technology that reach beyond the discipline and the academy. On a critical level, knowing whether a tool promotes disciplinary values and aims entails knowing something about *how* a tool works, beyond observing that it does.

It is surprisingly easy to forget how ineffectual search engines were in 1996 when Brin and Page began the project that would become Google. The Web was growing rapidly, and most search engines at that time used approaches that were easily exploited by commercial interests looking to game the system. What set Google apart was its use of a feature called PageRank. Treating links as analogous to academic citations—as “votes” for the value of a particular page (Brin and Page 1998)—and inventing a way to track links backward (something that had not been done before), Brin and Page were able to leverage the Web's own structure as a means of sorting information. The result was to immediately and dramatically improve the relevance of search results.

Today we take more or less for granted that a Web search will return relevant information, and programmers have developed more ways to customize an engine's search results; Google's search algorithm now takes into account more than 200 “signals,” or types of data, as it processes a search query (Levy 2011, 49). PageRank is one of these signals; another is the geographical location of the searcher's computer as detected from its IP address (“Location” 2011). The literal position of the researcher, in other words, can influence the search results. When I search for “coffee shops,” Google returns links for two national coffee chains, followed by a nearly complete list of coffee shops within walking or biking distance of my apartment. Location can affect other kinds of searches, as well. Technology writer

Steven Levy offers the following example, which shows how location can be used to help the search engine know how to interpret an ambiguous query: “Consider the query ‘houston baker.’ Was the user looking for a person who baked bread in Texas? Probably. But if you were making that query very far from the Lone Star State, it’s more likely that you were seeking someone named after the famous Texan” (2011, 50). Physical location and my choice of search terms are only a few factors that inform the outcome of a search, however. When I try my “coffee shop” search again—from the same location, but this time using a borrowed laptop with a very different browser history—the result now includes a list of cafés in a city two hours away.

Another example of how Google mediates the search experience is through its autocomplete feature, which displays a list of possible search phrases to users as they begin to type within the browser search box. These “predictions,” as Google calls them, are based on “a mix of popular searches by users around the world” and the user’s own Web history, if it is enabled (“Search Preferences” 2011). Here, Google draws from a store of collective and individual memory, its dynamic response inviting the searcher to engage in a microdialogue (“Did I mean to type Frances Mayes? Or Frances Maule? And who is Frances Maule Bjorkman?”) as we are presented with possibilities culled from the crowd and from our own past practices. These examples help show how researchers and technologies shape search outcomes in ways that cannot be accounted for by looking at researcher positionality independent of the technology.

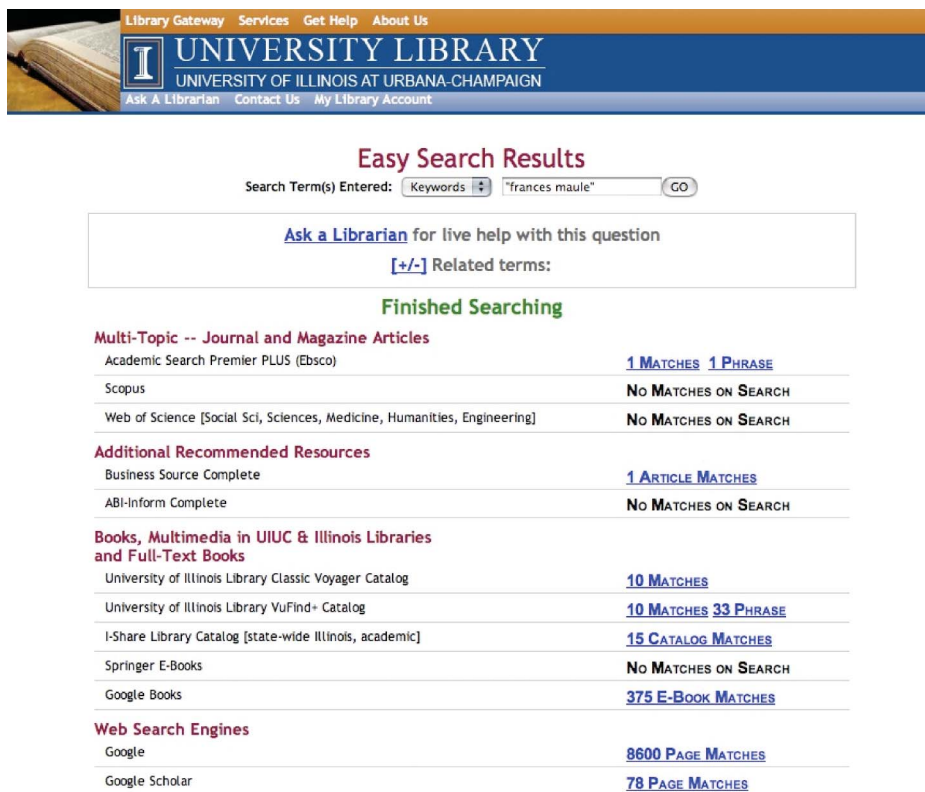
Electronic library catalogs have offered a different kind of search experience. If Google’s design exploits the wisdom of crowds and the linked structure of the Web, library catalogs (and similar sorts of scholarly databases) have often presented users with a closed system founded on hierarchy, classification, and descriptions written by experts. Both search engines and electronic library catalogs employ databases and metadata, but users interact with that metadata in a library catalog in ways that renders its structuring function more visible: namely, as specific search categories like a text’s title, author, or subject headings that structure the available search possibilities. These metadata, usually created by a professional librarian or indexing service, are what make it possible for me to find items in the catalog’s database—assuming, of course, that I know or can figure out the right bits of descriptive information (subject headings, for example) to use in my search. It is this last part—intuiting the terms that will lead to useful sources—that is one challenge and limitation of catalog searches. As Wendy Sharer notes, “Description and indexing practices [help to] establish and perpetuate cultural and social values by allowing only certain materials to become visible to researchers, while obscuring others” (1999, 128). The artifacts that serve as bases for revisionist histories of the kind that interest Sharer are unlikely to be cataloged under the subject headings “composition” and “rhetoric.” To navigate these systems thus requires creative research

strategies such as “synonym generation, chaining, name collection, pearl growing, and successive segmentation” (Yakel 2010, 113–114). Because of these challenges, metadata is of particular importance to humanities scholars (Borgman 2009, 41; Nunberg 2009). Research environments that use metadata standards like the Dublin Core⁹ are crucial not just for supporting older research tools but also for fostering new approaches to search and interpretation that rely on accurate descriptive data and interoperability across systems.

Insofar as keywords really do act as “keys” to entry, we might think of catalog searches as encouraging entry through the front door—the more formal means of entry opened by a master key (a controlled vocabulary of authorized terms)—and full-text, metadata-rich searches as allowing entry through whatever openings we can squeeze in through (back door, bedroom window, cellar door). As I researched Frances Maule, I moved back and forth between both kinds of searches, as I expect most scholars now do. While each offered different means of discovery, I found the flexibility of the full-text search—which allowed me to enter a text through the back door or, as linguist Geoffrey Nunberg (2009) puts it, to “barrel into it sideways”—to be crucial in generating a list of sources whose treatment of Maule was otherwise not visible in the library’s online catalog. I increasingly found myself thinking in terms of keywords as I read and wrote. When Robert Connors famously remarked two decades ago that “search is play,” the scene he described was a researcher confronting an “inert, dusty mass of past records” (1992, 23). Mass digitization and other recent technological developments not only change the scene of play (and eliminate some of the dust), they also represent opportunities for *new kinds* of play. For some digital humanities scholars, these forms of play involve “big data” and sophisticated visualization programs,¹⁰ but play can be transformed by an ordinary Web browser as well. I found that Web-based searching allowed me not only to thicken my rhetorical scene more quickly but also to rapidly test and refine questions and hypotheses. Our scholarly rhythms and habits have begun to shift as we come to increasingly rely on the infrastructure of digital, networked technologies for our research—much as people’s behaviors, language, and daily rhythms shifted as they came to depend on electricity and its infrastructures (Star and Ruhleder 1996, 113).

My discussion thus far has divided digital searching into two general types—one more associated with libraries as closed search environments and another with Google’s search of the open Web—but in practice these are converging. Google Books is a case in point: one of Google’s main selling points for its “digital library” has been that the contents of the books it scans would be fully searchable, but it has applied library-style metadata to the scanned works, as well. In a piece for the *Chronicle of Higher Education*, Geoffrey Nunberg lambasts Google for its handling of this metadata, which he describes as a “train wreck” (2009). Citing widespread errors in publishing dates (e.g., 1899 for Stephen King’s novel *Christine*

and 1900 for a book of Virginia Woolf's letters) and errors in classification (Susan Bordo's *Unbearable Weight: Feminism, Western Culture, and the Body* was listed under "Health & Fitness"), Nunberg furthermore criticizes Google's decision to use the BISAC cataloging system, which he describes as "well-suited for a chain bookstore or a small public library" but a poor system for describing the aggregate holdings of large academic research libraries.¹¹ Fueling Nunberg's ire is his belief that no entity will be able to replicate or compete with the Google Books project—this is what we will have to work with for a good long while—and so it is particularly important that it be done right (which is to say, that it be accurate and able to support the work that scholars do). His point is well taken: Not only do many scholars rely on the tool in informal ways but some libraries have also begun to incorporate Google Books results into their own institutional interfaces. The University of Illinois at Urbana-Champaign, for example, displays results for Google, Google Scholar, and Google Books through its library's default "Multi-Subject" search (see Figure 1), and WorldCat has similarly incorporated links to Google Books into its interface.



The screenshot shows the University of Illinois at Urbana-Champaign Library Gateway search results for the query "frances maule". The page features a navigation bar with links to Library Gateway, Services, Get Help, and About Us. Below the navigation bar is a search bar with the text "Search Term(s) Entered: Keywords [frances maule] GO". A link to "Ask a Librarian" is provided for live help. The search results are categorized into several sections: "Multi-Topic -- Journal and Magazine Articles" (1 MATCHES 1 PHRASE), "Additional Recommended Resources" (1 ARTICLE MATCHES), "Books, Multimedia in UIUC & Illinois Libraries and Full-Text Books" (10 MATCHES, 10 MATCHES 33 PHRASE, 15 CATALOG MATCHES, No MATCHES ON SEARCH, 375 E-BOOK MATCHES), and "Web Search Engines" (8600 PAGE MATCHES, 78 PAGE MATCHES).

Search Term(s) Entered:	Keywords	GO
frances maule		
Ask a Librarian for live help with this question		
[+/-] Related terms:		
Finished Searching		
Multi-Topic -- Journal and Magazine Articles		
Academic Search Premier PLUS (Ebsco)	1 MATCHES 1 PHRASE	
Scopus	NO MATCHES ON SEARCH	
Web of Science [Social Sci, Sciences, Medicine, Humanities, Engineering]	NO MATCHES ON SEARCH	
Additional Recommended Resources		
Business Source Complete	1 ARTICLE MATCHES	
ABI-Inform Complete	NO MATCHES ON SEARCH	
Books, Multimedia in UIUC & Illinois Libraries and Full-Text Books		
University of Illinois Library Classic Voyager Catalog	10 MATCHES	
University of Illinois Library VuFind+ Catalog	10 MATCHES 33 PHRASE	
I-Share Library Catalog [state-wide Illinois, academic]	15 CATALOG MATCHES	
Springer E-Books	NO MATCHES ON SEARCH	
Google Books	375 E-BOOK MATCHES	
Web Search Engines		
Google	8600 PAGE MATCHES	
Google Scholar	78 PAGE MATCHES	

FIGURE 1 Search Results for "Frances Maule" (December 30, 2011). (Color figure available online).

As a snapshot of converging and emergent research technologies, these examples offer only a glimpse of what is yet to come. Already in place (or close on the horizon) are features such as catalog entries offering text-to-phone options or QR (Quick Response) codes for smartphones; crowdsourced tagging, translation, or correction of materials in digitized collections; and interfaces (at the New York Public Library and in Harvard's ShelfLife and LibraryCloud project, for example) that allow library patrons to create and browse virtual bookshelves online.

I bring up Nunberg's critique here for a number of reasons, not least of which is that it points to a few of the structural pitfalls and challenges of digital searching. These challenges are located in the technology and its design as well as in the way it is taken up and used by researchers. At the same time that we recognize the potential of digital technologies, then, we must also keep in view their structural biases and limitations for the scholarly work we deem important. Digital search environments, like conventional physical archives, are built on "material acts of power" that "shape the body of evidence available to historians" (Sharer 1999, 121). Even as we are able to search in more ways and new ways for alternative rhetorics, it is important that we not let digital plentitude cloud our recognition of what and who is still excluded—or what and who may be newly erased.

RESEARCHER POSITIONALITY, PROXIMITY, AND SPEED

If we accept that critically reflecting on our research tools and technologies is a worthy and timely goal for historians, then how might we link that reflexivity to existing theoretical and methodological values? I see questions about technology's shaping effects as a logical extension of theoretical commitments to materiality and positionality, particularly as articulated by feminist scholars and woven into feminist research methods and methodologies (Collins 1999; Kirsch and Ritchie 1995; Kirsch and Royster 2010; Schell and Rawson 2010). An important contribution of feminist theory has been to trouble the ideal of the "objective" researcher and to insist that the researcher's methods and subject positions be named and made visible. Heidi McKee and James Porter offer a list of qualities they identify with the feminist/ethical researcher; three qualities from this list are especially germane to my discussion of technology and positionality: (1) "self reflexivity and critical consciousness about one's own position, gender, and status," (2) willingness to "consult and consider a variety of viewpoints," and (3) a commitment to transparency: "to making the process and constructed nature of research visible" (2010, 155–156). These values speak to some of the potential benefits of digital technologies (for recovering marginalized rhetorics and generating thick description), as well as the need to use them in self-conscious ways (understanding how their structures might transmit

values or perpetuate omissions). To bring a feminist sensibility to historical research is thus to demand a closer consideration of the technologies that mediate our access to, and interaction with, the raw materials of historical memory.

In service of this goal, I will briefly outline the principle of “proximity” as a descriptive device and heuristic for critically reflecting on research processes. This principle draws from the idea that researchers tend to choose research subjects that are in one way or another “close” to them. This closeness is threaded through existing scholarship—from the observation that we may be more drawn to research subjects we like, admire, or identify with; to the idea that our research questions emerge from present-day teaching practices; to the fact that the burdens of travel to distant archives may make some subjects or lines of inquiry more appealing while discouraging or curtailing others. I introduce three forms of proximity—*affective*, *geographical*, and *virtual*—which map onto existing narratives of historical research and which foreground links between (1) identity- or ideology-oriented notions of researcher positionality and (2) material and technological elements of the research process.

The first of these three forms of proximity is *affective*: the intellectual and emotional¹² investments and orientations that drive a researcher’s choice of topic and affect the way she pursues those topics. Carol Mattingly, for example, argues that researchers may gravitate toward research subjects who align with present-day feminist sensibilities (2002, 101–102); Jacqueline Jones Royster argues that researchers should declare their “passionate attachments” (280); and Connors writes of the way current-day teaching practices or questions can form the basis of an archival inquiry (1992, 22). These affective investments converge around the familiar idea that we “bring to the archive” certain values, interests, identifications, or experiences. Whether they come from what is close to our hearts or on our minds, these inclinations operate as terministic screens in our identification and selection of research topics.¹³

The second form is *geographical* proximity, which describes the physical location of the materials in relation to the researcher or otherwise accounts for the influence of geography and location on the research process (for examples of the latter, see Kirsch and Rohan 2008, particularly essays by Gesa Kirsch, W. Ralph Eubanks, and Malea Powell). Asking about location—where do researchers go, how do they search and engage the archive, and in what spaces?—serves to remind that research is an embodied activity, whether online or off. And, as the earlier discussion of Google’s search algorithm and the “coffee shop” search suggests, even online searches can be affected by the physical location of that embodied activity.

To these first two forms of proximity, we can add a third—*virtual* proximity—which describes the potential to find or encounter a source through the use of finding aids, search technologies, metadata, and similar

mechanisms. This form captures the sense of virtual “nearness” to sources or ideas—particularly as that nearness is enabled by organization, search, and retrieval technologies. The more visible or discoverable the source, we might say, the more “proximate” it is. Proximity in this sense could be a measure of the searchability of a digital object, an indication of the density of metadata connected to that object, or a function of how the Google search algorithm ranks the object in its results (based on the searches of other users, for example). In digital environments, proximity will thus depend upon any number of variables, including search tools, disciplinary norms and methodologies, and the skills and proclivities of the researcher.

My interest in a category like virtual proximity involves reframing research technologies that have become mainstream, like Web-based searches, in order that we might consider ways of weaving these tools into our methodologies in more reflexive and critical (read: not merely instrumental) ways. Although proximity is a spatial concept, its effects are perhaps most apparent in terms of time or speed. I have encountered a few scholars who seem to regard digital search technologies as either a threat to good research or as effecting a mere “speeding up” of the research we’ve always done—as though the speed associated with those time-saving tools functioned along a single vector, such that the speeding up could occur without meaningfully altering the aim or outcome of the research project, or by extension reshaping the larger scholarly, disciplinary enterprise of which that project is a part. I hope it is clear by now that I do not agree with either of these positions (though the former probably deserves a fuller discussion that I can give here).

Even when technology speeds up parts of the research process, it does not mean the time it actually takes to complete a book or article is necessarily reduced. Technology may enable a different, more ambitious framing or scale of inquiry than would otherwise have been possible or realistic to pursue. As barriers to accessing primary sources are lowered, digital tools may result in a devaluation of certain kinds of historical work or certain parts of the research process (Stallybrass 2007; Bowker and Star 2000, 109). Or it might result in “research inflation”—much as the development of household technologies in the mid-twentieth century simply raised expectations of what housewives were expected to accomplish. Nevertheless, I would suggest that digital spaces also provide cheap, safe spaces to try out ideas. In this context, speed creates an environment conducive to experimentation by lowering the stakes of trying out a risky idea, or testing a hunch, or pursuing a seemingly intractable question or subject. Those stakes might take the form of time, money, or the emotional investment involved in testing an idea. “Speed” refers not only to the quickness with which a library database returns search results but to time that is spent differently. The digital search doesn’t simply speed up our “predigital” search methods—it shifts and transforms the epistemological spaces we

occupy as researchers. It creates new habits, new ways of interacting with information, and new opportunities for serendipity as we move through texts.

As a temporal-spatial concept, proximity invites us to consider research as situated in space, within a larger context or scene of activity. It is also a *relational* concept and so denotes relationships within that scene: between researcher and artifact, researcher and imagined subject, and the researcher's evolving understanding in response to the "data" gathered. The influence of research methods and methodologies can be traced across all three types of proximity. For example, the value that some scholars have come to place in "local" histories, "little narratives," and thick description (Miller 1993, Daniell 1999) mediate data-gathering practices as well as interpretive ones. In underscoring the spatial and relational structures of historical research, proximity might serve as a reminder that researcher positionality is technologically mediated as well as ideologically constituted. Crucially, while principles of proximity do not prioritize digital technologies, they do create an epistemological space within which to read and analyze technologies and research practices as mutually shaping; they invite us, as well, to consider both digital and nondigital technologies, which are often so enmeshed that it makes little sense to treat "digital methods" as something that can be cordoned off from the general work of historical research.

Even research that makes use of traditional archives is increasingly likely to be mediated by digital technology. During a recent trip to the Sophia Smith Collection and Smith College Archives, I asked the archivist if she knew of any collections that might contain information about Frances Maule. She led me first to the archive's (wooden) card catalog, then to a computer where she ran searches across an internal database, then to a cabinet of hanging file folders with copies of paper finding aids, and finally offered to run a full-text search across a digital folder that held copies of all finding aids as searchable Word documents. Later, I used my digital camera (as many visitors to archives now do) to take high-resolution photos of documents so that I could add them to the files on my laptop: the site of my own digital, searchable, portable archive. If I had been inclined to reinvent my note-taking habits, I might have entered citations and notes for my newfound materials using Mendeley or Zotero, but I am digitally wedded to Microsoft Word by the sheer volume of research material I have committed to that form. I see this accumulation and mixing of old and new technologies, old and new forms and formats, as a further impetus to reflect on research technologies more broadly. New technologies build upon or remediate older ones. Older, dustier technologies seem "new" to the newly initiated.

As search boxes become ever more ubiquitous and familiar, we should not mistake familiarity for an understanding of underlying structures or an awareness of the shaping forces at work on our research processes. One aim

of critical reflection, then, is to make the actual processes by which we do research more readily subject to our collective disciplinary gaze. Proximity, as I have presented it here, is certainly not the only way to do this. But because it cues us to physical spaces and relationships as well as figurative (or less tangible) ones, it has the advantage of bringing to our existing concern with positionality a more acute sense of research tools and technologies, without necessarily needing to divide those tools into categories of analog and digital, “old” and “new.”

CONCLUSION

I have argued in this article that one important way for historians in rhetoric and composition to bring critical agency to the practice of historical research is by reflecting on the role that digital technologies play in their own research processes. In closing, I want to turn (with a nod to the opening epigraphs from Enos [2009] and Drucker [2011]) to a second avenue for such critical agency—namely, through our capacity as rhetoricians to shape, imagine, and build new tools that offer support for, or innovative reimaginings of, our disciplinary goals and inquiries. If ever there was a *kairotic* moment for change, experimentation, and reinventing our research tools, then we are in it. The question is: What, if anything, are we going to do about it? In a 2010 report sponsored by the Council on Library and Information Resources, Charles Henry and Kathlin Smith remark:

It is not hard to intuit the potential for scholarly research of digitizing millions of volumes of books. Access to a sweeping array of our cultural heritage, in a form that can be queried, interpreted, and reconstituted as new knowledge, could transform scholarship. Yet to best support research, these digital databases must be organized and built to reflect the methodologies and intellectual strategies of contemporary scholarship. Until recently, discussions of large-scale projects have tended to focus on copyright, technical, and quality issues; *few have included the perspectives of scholars on how they use these resources*. (106, my emphasis)

The idea that scholars have not had much of a role in shaping these emerging research tools and environments is troubling. Technology affects us all and should be a concern for every scholar. These questions should not be left solely to technologists, librarians, or only those scholars who self-identify with the digital humanities, though there are people in all three areas doing creative and thoughtful work. People in the aforementioned groups can be viewed as “the leading edge of a transformational wave” but, as Andrew Stauffer puts it to his colleagues in Victorian studies, “We all have

a stake in the forms of research, publication, use, archiving, and professional credit that are taking shape in the digital environment. . . . How we navigate those changes . . . depends on a broad-based conversation about values, practices, and goals that we all share as readers, students, professors, and custodians of the nineteenth-century archive” (2011, 295–296). Rhetoric and composition historians cannot so easily unite around a single historical era, of course, but critical reflections directed at our own current research practices and uses of technology offer one way to help build toward the sort of conversation Stauffer and others call for, while keeping it grounded in our own disciplinary projects and values.

In a March 2011 blog post, digital historian Dan Cohen¹⁴ described being at a gathering of technologists and scholars where the topic under discussion was “what a large digital library should look like.” He described technologists gathered on one side of the room discussing things like “databases, indices, and search mechanisms” while on the other side of the room the scholars were imagining what the digital archive’s *interface* would look like. Their idea was that it should be “as close to the library as possible, where one could pull down boxes from the shelves, and then open those boxes and leaf through the folios one by one.”

I close with this anecdote for two reasons. First, because it highlights a tendency to identify historical research with labor that occurs in certain archetypal settings (in dim basements, at long, wooden library tables, seated near rolling carts stacked with archival boxes). This is an image that not only captures the material conditions of (some kinds of) archival work but also serves to remind that the cachet of scholarly/archival work has been maintained, in part, “through its privileged relation to the protection and retrieval of scarce resources” (Stallybrass 2007, 1581–1582). The promise of mass digitization can thus be seen to represent, on the one hand, a widening of access and discovery and new means to recover marginalized or forgotten rhetorics and, on the other, a possible recalibration of the criteria by which historical scholarship is measured. Embracing these technologies can help rhetoric and composition scholars pursue the goal of recovery work for revisionist histories, but it may also change the way that our work is valued and result in altered expectations, if it has not begun to do so already. But there is a second reference suggested by Cohen’s anecdote, and it is with this more hopeful reference that I want to end: I see the collaboration of scholars and technologists as a reminder that the project of history is about building and creating as well as excavating—about looking forward as well as back. If we are to be alert to the potentials and pitfalls of the digital archive, and if we hope to be participants in meaningful dialogue about its future development, then we must be willing to reflect on how we use and are shaped by our research technologies—as they relate to the needs and values of the discipline, to the structures of academic libraries and archives, and to the wider information-communication environment.

NOTES

1. This number has been called into question by some critics. See Stokes (2010).
2. I mean physical reorientation in the quite literal sense: Digital research allows us to undertake types of inquiry while sitting before our computers that used to happen by other means. Although it's not possible to peer into our own brains, Nicholas Carr (2010) further suggests that routine activities (such as reading and conducting research in digital environments) may reshape the anatomy, the wiring, of the brain itself.
3. I am particularly thinking of Bruno Latour, for whom objects can be "participants in the course of action," and social action can be "shifted or delegated to different types of [non-human] actors which are able to transport the action further" (2005, 70–71).
4. The use of computing in libraries and scholarly research is not a recent event; libraries have made use of databases since at least the 1960s or 1970s, and more routinely since the 1980s when the development of the microcomputer made it easier to make these databases more widely available to students and faculty. It has been only in the last decade, however, that developments in the area of mass digitization have enabled digital search on such an unprecedented scale. I will not attempt here to rehearse the many conversations about technology that have unfolded in the past 25 or so years, which we can learn from but which have tended to happen in other forums or other specializations, like those previously mentioned as well as library and information sciences, technical communication, and new media and hypertext studies.
5. More recently, Kirsch and Royster used the term *strategic contemplation* to similarly describe scholars' reflections on their own research as a "lived process" (2010, 656–657).
6. The four books are *She Strives to Conquer* (1937, reprint of 1934 edition), *The Road to Anywhere* (1938), *Girl with a Pay Check* (1941), and *Careers for the Home Economist* (1943).
7. Heterodoxy was a radical New York–based luncheon and debate club for women whose members were mainly educated professional women, many of whom lived in New York City's Greenwich Village. The club was founded in 1912 by the suffragist and former-Unitarian minister Marie Jenney Howe, who chose the name "Heterodoxy" because the only requirement for being a member was that one's thinking not be orthodox.
8. As Latour explains, "black box" is a term "used by cyberneticians whenever a piece of machinery or a set of commands is too complex" (1987, 2–3). Rather than represent that complexity in all of its detail, they "draw a little box about which they need to know nothing but its input and output" (1987, 3). For Latour, a black box can be any technology or idea whose original complexities are covered over by familiarity and acceptance.
9. More information about the Dublin Core Metadata Initiative (DCMI) can be found at <http://dublincore.org>.
10. Two examples of well-known projects that involve data visualization are the Republic of Letters project at Stanford, which is suggestive of ways that we might think about studying circulation (<https://republicofletters.stanford.edu>), and the Google Ngram Viewer (<http://books.google.com/ngrams>) and culturomics projects (Bohannon 2011). For an introduction to "big data" projects in English studies, see Hayles (2009), particularly her discussion of "reading" and scale.
11. An earlier version of Nunberg's *Chronicle* article is posted at his own blog; the reader comments there are enlightening, particularly the lengthy response posted by Jon Orwant, manager of the Google Books metadata team.
12. Bizzell's discussion of emotion in feminist historical research (2000, 12) is relevant here.
13. I use terms like *identify* and *select* advisedly here: part of what my own experience with Frances Maule illustrates is that processes of identifying and selecting may require, and thus be distributed across, numerous interactions with the archive. In one sense, my "selection" of Maule happened when I found the 1944 bibliography and decided to request one of her books; in another, it happened when I finally decided I had enough information to make her worth writing about.
14. Dan Cohen is director of the Center for History and New Media at George Mason University; he is well known for his work in digital history and digital humanities, which includes books (e.g., *Digital History*), articles, blogging, podcasts, and the development of digital tools for humanists (e.g., Omeka, Zotero).

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